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Practic Questions:

Lec # 23

① Methods of presentation of data?

There is many method of presentation of a data:

(i) Text (ii) Data in rows

(iii) Tabular (iv) Graphic (v) charts and graphs

2) Difference b/w line graph and sector:

Line Graph:

A line graph plots data as points and then join them with a line.

∴ "A graphs in which point join with a line called line graph." (Umaid)

Sector graph:

Sector graphs use the division of a circle into different sector.

Lec # 24

① Find Mean, media, mode:

Sol.FOR Mean = ?

X
13
18
13
14
13
16
14
21
13
135

$$\text{Mean} = ?$$

$$\bar{x} = \frac{\sum x}{n}$$

$$\bar{x} = \frac{135}{9}$$

$$\bar{x} = 15$$

$$\boxed{\text{Mean} = 15}$$

For Median

Arrange the data:

$$= 13, 13, 13, 13, 14, 14, 16, 18, 21$$

The middlest value is median

That is 14

$$\boxed{\text{Median} = 14}$$

For modeThe most repeated value
is mode

That is 13

$$\boxed{\text{mode} = 13}$$

Find range, class width, relative frequency:

Sol.

X	36, 48, 54, 92, 57, 63, 66, 76, 66, 80
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Range:

$$\text{Range} = \text{max. value} - \text{min. value}$$

$$\text{Range} = 92 - 36$$

$$\text{Range} = 56$$

Class width:

$$\text{class width} = \frac{\text{Range}}{\text{No. of class } = n}$$

$$= \frac{56}{10} = 5.6 \Rightarrow 6$$

∴ ★ ∴

Lec #25

①

Find Mean:

Sol.

Distance class/	f	fx	X
1-10	10	55	5.5
11-20	20	310	15.5
21-30	40	1020	25.5
31-40	30	1065	35.5
	100	2450	

Mean for grouped data

$$\bar{x} = \frac{\sum fx}{\sum f}$$

$$\bar{x} = \frac{2450}{100}$$

$$\bar{x} = 24.5$$

$$\boxed{\text{Mean} = 24.5}$$

Question # 2Find Arithmetic mean:Sol, $N = 20$

X	80, 75, 90, 69, 83, 55, 79, 93, 58, 94, 89
---	--

A.M = ?

$$\bar{x} = \frac{\sum x}{N}$$

$$\bar{x} = \frac{865}{20}$$

$$\bar{x} = 43.25$$

This Arithmetic is found

By population.

$$\therefore \bar{x} = \frac{\sum x}{N}$$

 $N \rightarrow$ population

① Find mean using empirical relationship:
sol.

Given that:

$$\text{Median} = 15, \text{ Mode} = 22$$

We know that, empirical relationship is:

$$\text{Mean} - \text{Mode} = 3 (\text{Mean} - \text{Median})$$

$$\text{Mean} - \text{Mode} = 3 \text{ Mean} - 3 \text{ Median}$$

$$\Rightarrow 3 \text{ Mean} - \text{Mean} = 3 \text{ Median} - \text{Mode}$$

$$2 \text{ Mean} = 3(15) - 22$$

$$2 \text{ Mean} = 45 - 22$$

$$2 \text{ Mean} = 23$$

$$\text{Mean} = 23/2$$

$$\boxed{\text{Mean} = 11.5}$$

Question No. 2

Find G.M and H.M:

sol.

Given that is:

$$80, 85, 90, 95, 100$$

For G.M

$$\text{suppose } x_i = 80, 85, 90, 95, 100$$

$$\text{G.M} = \sqrt[5]{x_1 \cdot x_2 \cdot x_3 \cdot \dots \cdot x_5}$$

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$$G.M = \sqrt[5]{80 \times 85 \times 90 \times 95 \times 100}$$

$$G.M = \sqrt[5]{5814000000}$$

For H.M

Given that

suppose $x_i = 80, 85, 90, 95, 100$

$$H.M = \frac{n}{\sum \left(\frac{1}{x_i}\right)}$$

$$H.M = \frac{5}{\left(\frac{1}{80} + \frac{1}{85} + \frac{1}{90} + \frac{1}{95} + \frac{1}{100}\right)}$$

X	$\frac{1}{x}$
80	0.0125
85	0.0118
90	0.0111
95	0.0105
100	0.01
	0.0559

$$H.M = \frac{5}{(0.0125 + 0.0118 + 0.0111 + 0.0105 + 0.01)}$$

$$H.M = \frac{5}{0.0559}$$

$$H.M = 90.091$$

Sol.

The data is given that:

(27, 28, 30, 32, 34, 38, 41, 42, 44, 44, 46, 53, 56, 62)

For symmetrical distribut
first we will find mean, median
and mode:

Mean

$$\bar{x} = \frac{\sum x}{n}$$

$$\bar{x} = \frac{577}{14}$$

$$\bar{x} = 41.21$$

72

MedianFor Even

$$\begin{aligned}\text{Median} &= \frac{\text{next } 34+38}{2} \\ &= 72/2 \\ &= 36\end{aligned}$$

For Mode:

Most repeated value

$$\text{mode} = 44$$

① Find Quartiles:

Sol:

Given that:

Arranged) 18, 20, 23, 20, 23, 27, 24, 23, 29

18

20

20

23

23

23

24

27

29

For $Q_1 \Rightarrow \frac{n(n+1)}{4}$

$$= \frac{1(9+1)}{4} = \frac{10}{4} = 2.5$$

So,

$$Q_1 = 2^{\text{nd}} \text{ value} + 0.5 (3^{\text{rd}} \text{ value} - 2^{\text{nd}} \text{ value})$$

$$= 20 + 0.5 (20 - 20)$$

$$= 20 + 0.5 (0)$$

$$Q_1 = 20 + 0 = 20$$

For $Q_2 \Rightarrow \frac{n(n+1)}{4}$

$$= \frac{2(9+1)}{4} = \frac{2(10)}{4} = \frac{20}{4} = 5$$

So,

$$Q_2 = 5^{\text{th}} \text{ value} + 0.0 (6^{\text{th}} \text{ value} - 5^{\text{th}} \text{ value})$$

$$Q_2 = 23$$

For $Q_3 \Rightarrow \frac{n(n+1)}{4}$

$$= \frac{3(9+1)}{4} = \frac{3(10)}{4} = \frac{30}{4} = 7.5$$

So,

$$\begin{aligned}
 &= 7^{\text{th}} \text{ value} + 0.5 (8^{\text{th}} \text{ value} - 7^{\text{th}} \text{ value}) \\
 &= 24 + 0.5 (27 - 24) \\
 &= 24 + 0.5 (3) \\
 &= 24 + 1.5 \\
 &= 25.5
 \end{aligned}$$

$$\boxed{Q_3 = 25.5}$$

Question No. 2

Find mid hinge and Quartile Deviation.

Sol.

Given data is:

4, 17, 7, 14, 18, 12, 3, 16, 10, 4, 4, 11

Arranged the data: 17, 18

3, 4, 4, 4, 7, 10, 11, 12, 14, 16, 17, 18

→ 3, 4, 4, 4, 7, 10, 11, 12, 14, 16, 17, 18

Now,

Mid hinge:

$$\text{Mid hinge} = \frac{Q_1 + Q_3}{2}$$

$$Q_1 = \frac{n(n+1)}{4} = \frac{1(12+1)}{4} = \frac{13}{4} = 3.25$$

So,

$$Q = 3^{\text{rd}} \text{ value} + 0.5 (4^{\text{th}} \text{ value} - 3^{\text{rd}} \text{ value})$$

$$= 4 + 0.5(4-4)$$

$$= 4 + 0.5(0)$$

$$Q_1 = 4$$

$$Q_3 = \frac{n(n+1)}{4}$$

$$= \frac{3(12+1)}{4} = \frac{3(13)}{4} = \frac{39}{4} = 9.75$$

So,

$$Q_3 = 9^{\text{th}} \text{ value} + 0.75(10^{\text{th}} \text{ value} - 9^{\text{th}} \text{ value})$$

$$= 14 + 0.75(16 - 14)$$

$$= 14 + 0.75(2)$$

$$= 14 + 1.5$$

$$Q_3 = 15.5$$

Then,

$$\text{Mid hinge} = \frac{Q_1 + Q_3}{2}$$

$$= \frac{4 + 15.5}{2}$$

$$= \frac{19.5}{2}$$

$$= 9.75$$

Solved By:

BS Mathematic

Umar-Abbasi

Quartile Deviation:

$$Q.D = \frac{Q_3 - Q_1}{2}$$

We have finded that:

$$Q_1 = 4$$

$$Q_3 = 15.5$$

So,

$$Q.D = \frac{15.5 - 4}{2}$$

$$Q.D = 11.5 / 2$$

$$Q.D = 5.75$$

Question # 3

Find Range and mid Range:

Sol.

The Given data is:

14, 10, 17, 5, 9, 20, 8, 24, 22

We arrange the data:

~~5, 9, 8, 10,~~

5, 8, 9, 10, 14, 17, 20, 22, 24

Range:

Range = Max. value - Mini. value

$$= 24 - 5$$

$$\text{Range} = 19$$

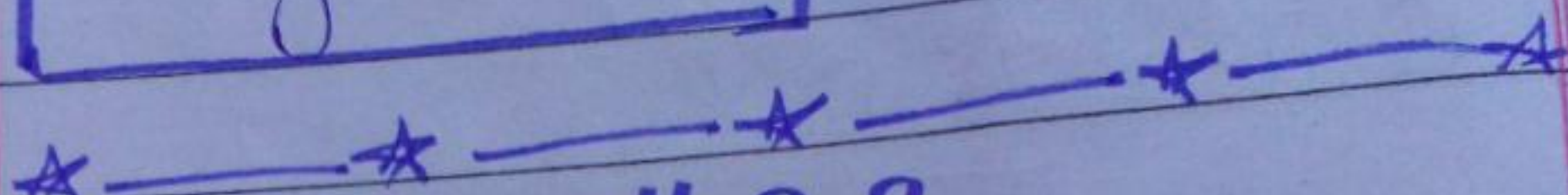
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Mid Range:

$$\text{Mid Range} = \frac{\text{Large value} + \text{Small value}}{2}$$

$$= \frac{24 + 5}{2} = 29/2$$

$$\text{Midrange} = 14.5$$



Lec # 28

Find Standard Deviation.

Sol.

Given that:

Standard Deviation of first ten natural number (1-10).

X_i	$X_i - \bar{X}$	$(X_i - \bar{X})^2$
1	$1 - 5.5 = -4.5$	20.25
2	$2 - 5.5 = -3.5$	12.25
3	$3 - 5.5 = -2.5$	6.25
4	$4 - 5.5 = -1.5$	2.25
5	$5 - 5.5 = -0.5$	0.25
6	$6 - 5.5 = 0.5$	0.25
7	$7 - 5.5 = 1.5$	2.25
8	$8 - 5.5 = 2.5$	6.25
9	$9 - 5.5 = 3.5$	12.25
10	$10 - 5.5 = 4.5$	20.25
55		82.5

$$S.D = \sqrt{\frac{\sum (X_i - \bar{X})^2}{N}}$$

$$\mu = \bar{X} \quad N = 10$$

$$\bar{X} = ?$$

$$\bar{X} = \frac{\sum X}{n}$$

$$\bar{X} = 55/10$$

$$\bar{X} = 5.5$$

$$S.D = \sqrt{\frac{\sum (X_i - \mu)^2}{N}}$$

$$S.D = \sqrt{\frac{82.5}{10}}$$

$$= \sqrt{8.25}$$

$$S.D = 2.872$$

Question No. 2

Find Variance and
Coefficient of Variance: = ?

Sol.

Given that:

X_i	$X_i - \bar{X}$	$(X_i - \bar{X})^2$
5	$5 - 4.5 = 0.5$	0.25
6	$6 - 4.5 = 1.5$	2.25
2	$2 - 4.5 = -2.5$	6.25
3	$3 - 4.5 = -1.5$	2.25
1	$1 - 4.5 = -3.5$	12.25
7	$7 - 4.5 = 2.5$	6.25
4	$4 - 4.5 = -0.5$	0.25
8	$8 - 4.5 = 3.5$	12.25
36.		42

"we use 'N' because the data is given from population"

$$\text{Var}(X) = \frac{\sum (x_i - \bar{x})^2}{N}$$

$$\mu = \bar{x} = ? = \frac{\sum x}{n}$$

$$= \frac{36}{8} = 4.5$$

$$\text{Var}(X) = \frac{\sum (x_i - \bar{x})^2}{N}$$

$$= \frac{42}{8} = 5.25$$

$$\boxed{\text{Var}(X) = 5.25}$$

Coefficient of Variance:

$$C.V = \frac{S}{\bar{x}} \times 100$$

"S" is standard Deviation.

We have $\text{Var}(X)$, So,

$$S = \sqrt{\text{Var}(X)}$$

$$S = \sqrt{5.25}$$

$$S = 2.291$$

Then,

$$C.V = \frac{S}{\bar{x}} \times 100$$

$$= \frac{2.291}{4.5} \times 100$$

$$= 0.5091 \times 100$$

$$\boxed{C.V = 50.9 \Rightarrow 51}$$

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Question # 3

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Find Mean Deviation.

Sol:

Given that:

X	$X_i - \bar{X}$	$ X_i - \bar{X} $
3	$3 - 2.85 = 0.15$	0.15
0	$0 - 2.85 = -2.85$	2.85
1	$1 - 2.85 = -1.85$	1.85
5	$5 - 2.85 = 2.15$	2.15
4	$4 - 2.85 = 1.15$	1.15
3	$3 - 2.85 = 0.15$	0.15
2	$2 - 2.85 = -0.85$	0.85
6	$6 - 2.85 = 3.15$	3.15
4	$4 - 2.85 = 1.15$	1.15
2	$2 - 2.85 = -0.85$	0.85
3	$3 - 2.85 = 0.15$	0.15
3	$3 - 2.85 = 0.15$	0.15
0	$0 - 2.85 = -2.85$	2.85
7	$7 - 2.85 = 4.15$	4.15
1	$1 - 2.85 = -1.85$	1.85
1	$1 - 2.85 = -1.85$	1.85
2	$2 - 2.85 = -0.85$	0.85
3	$3 - 2.85 = 0.15$	0.15
4	$4 - 2.85 = 1.15$	1.15
6	$6 - 2.85 = 3.15$	3.15
57		30.6

$$M.D = ?$$

M.D from Mean is:

$$M.D = \frac{\sum |X_i - \bar{X}|}{n}$$

$$\sum X = \frac{57}{20}$$

$$\bar{X} = 2.85$$

Now,

$$M.D = \frac{\sum |X_i - \bar{X}|}{n}$$

$$= \frac{30.6}{20}$$

$$M.D = 1.53$$

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Lec #29

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①

Find correlation coefficient:

Sol.

Given that

X	Y	XY	X ²	Y ²
1	9	9	1	81
2	6	12	4	36
4	4	16	16	16
6	12	72	36	144
7	8	56	49	64
10	3	30	100	9
30	42	195	206	350

correlation = ?

$$r = \frac{n \sum XY - \sum X \sum Y}{\sqrt{[n \sum X^2 - (\sum X)^2] \cdot [n \sum Y^2 - (\sum Y)^2]}}$$

$$r = \frac{6(195) - (30)(42)}{\sqrt{[6(206) - (30)^2][6(350) - (42)^2]}}$$

$$r = \frac{1170 - 1260}{\sqrt{(1236 - 900)(2100 - 1764)}}$$

$$r = \frac{-90}{\sqrt{(336)(336)}}$$

$$r = \frac{-90}{336} \Rightarrow r = -0.268$$

$$r = -0.268$$

② Find $\text{COV}(X, Y)$

Sol:

Given that

$$r = 0.5$$

$$\text{Var}(X) = 100$$

$$\text{Var}(Y) = 25$$

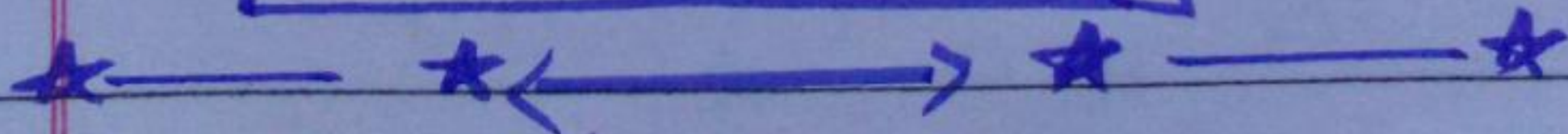
$$\text{COV}(X, Y) = ?$$

We know that:

$$r = \frac{\text{COV}(X, Y)}{\sqrt{\text{Var}(X) \cdot \text{Var}(Y)}}$$

$$\begin{aligned}\Rightarrow \text{COV}(X, Y) &= r \cdot \sqrt{\text{Var}(X) \cdot \text{Var}(Y)} \\ &= 0.5 \times \sqrt{100 \cdot 25} \\ &= 0.5 \times \sqrt{2500} \\ &= 0.5 \times 50\end{aligned}$$

$$\boxed{\text{COV}(X, Y) = 25}$$



Now is

ending:

Completed By:

(Jani Umar)

BS Mathematic)

Lec # 30

① Find least square regression

Line $y = a + bx$:

Sol.

Given that:

X_i	Y_i	$x_i y_i$	x_i^2
0	2	0	0
1	3	3	1
2	5	10	4
3	4	12	9
4	6	24	16
10	20	49	30

$$\hat{y} = a + bx$$

$$b = \frac{n \sum x_i y_i - (\sum x_i)(\sum y_i)}{n \sum x_i^2 - (\sum x_i)^2}$$

$$b = \frac{5(49) - (10)(20)}{5(30) - (10)^2}$$

$$b = \frac{245 - 200}{150 - 100}$$

$$b = \frac{45}{50}$$

$$b = 0.9$$

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$$a = \bar{y} - b\bar{x}$$

$$\bar{y} = \frac{\sum y_i}{n}$$

$$\bar{y} = 20/5 \Rightarrow 4$$

$$\bar{x} = \frac{\sum x_i}{n}$$

$$\bar{x} = 10/5 \Rightarrow 2$$

$$\therefore b = 0.9$$

$$\therefore a = \bar{y} - b\bar{x}$$

$$a = 4 - (0.9)(2)$$

$$a = 4 - 1.8$$

$$a = 2.2$$

Now,

$$\hat{y} = a + bx$$

$$\hat{y} = 2.2 + 0.9x$$

Question No. 2

Sol.

x_i	y_i
3	6
5	4
7	2
15	12

(Find the intercept?)
 $a = \bar{y} - b\bar{x}$

$$\bar{y} = \frac{\sum \bar{y}_i}{n}$$

$$\bar{y} = 12/3 = 4$$

$$\bar{x} = \frac{\sum x_i}{n}$$

$$\bar{x} = 15/3 = 5$$

Given that

$$b = 0.55$$

so,

$$a = \bar{y} - b\bar{x}$$

$$a = 4 - (0.55)(5)$$

$$a = 4 - 2.75$$

$$a = 1.25$$

$$\boxed{a = 1.25}$$

Question No. 3

find the slope
of the given

Data:

Sol.

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x_i	y_i	$x_i y_i$	x_i^2
2	6	12	4
10	7	70	100
4	3	12	16
8	4	32	64
24	20	126	184

$$b = \frac{n \sum x_i y_i - (\sum x_i)(\sum y_i)}{n \sum x_i^2 - (\sum x_i)^2}$$

$$b = \frac{4(126) - (24)(20)}{4(184) - (24)^2}$$

$$b = \frac{504 - 480}{736 - 576}$$

$$b = \frac{24}{160}$$

$$b = 0.15$$

That's

Good!